# Muscle Model · Teacher and Cover Plan

Learning goal: I can test a card and string arm model to show how two muscles pull a joint.

## Scheme Outcome

BASE Science scheme of work, Unit 1 (Skeletons, Muscles and Nutrition), Topic: Build and test a simple hinge model of an arm to show how muscles pull in pairs.

Use a card-and-string model of an arm joint to demonstrate and explain how paired muscles pull bones in opposite directions.

## Success Criteria

I can say which real body part each part of the model stands for.

I can pull String A and String B to move the card arm.

I can explain why pushing the string does not move the card bone.

## Model Sequence (I do)

First, check the parts: two stiff card strips are the bones, and the pin is the joint.

Next, pull top String A while String B is loose. The lower card strip bends up.

Then, pull back String B while String A is loose. The card strip goes straight again.

Our model is good because it shows why two strings are needed to pull both ways.

## Made by Matt

Our model also has limits. A real muscle gets fat and short when it pulls.

A piece of string stays the same length; your hand pulls the end of the string.

## We Do & Hinge Check

What happens if you pull both String A and String B hard at the same time?

Answer: A. If both pull at once, they pull against each other and the joint stays stuck.

## Safe Practical & Card Setup

Test the pre-made card and string arm model on your desk and record each test.

Kit: Pre-punched card strips, split pins with flat backs taped over, and two coloured strings (A and B).

Use pre-punched holes only. Keep split-pin ends folded flat under tape. Pull strings gently.

## You Do Routes (Supported, Independent, Aspire)

Supported: Match the three model parts (card, pin, string) to (bone, joint, muscle) and tick what happens when String A pulls. (Help: Card = bone. Pin = joint. String = muscle. Pull String A = arm bends up.)

Independent: Complete the 3-row model test table (Pull String A, Pull String B, Push String A) and write one sentence explaining why we need two strings. (Help: We need two strings because one string can only __, so String B must pull the arm __.)

Aspire: Explain one way the card model is like a real arm and one way a real muscle is different from a string. (Help: Both only pull in pairs, but a real muscle gets short by itself while a string needs __.)